

Smart Home Health Monitor

Final Presentation — CSC 494 IoT

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The Problem

Air purifiers lie about air quality.

Their built-in sensors are located **inside the device** — measuring air that has already been filtered.

The air quality at the purifier = **always clean**

The air quality where you breathe = **unknown**

Research shows:

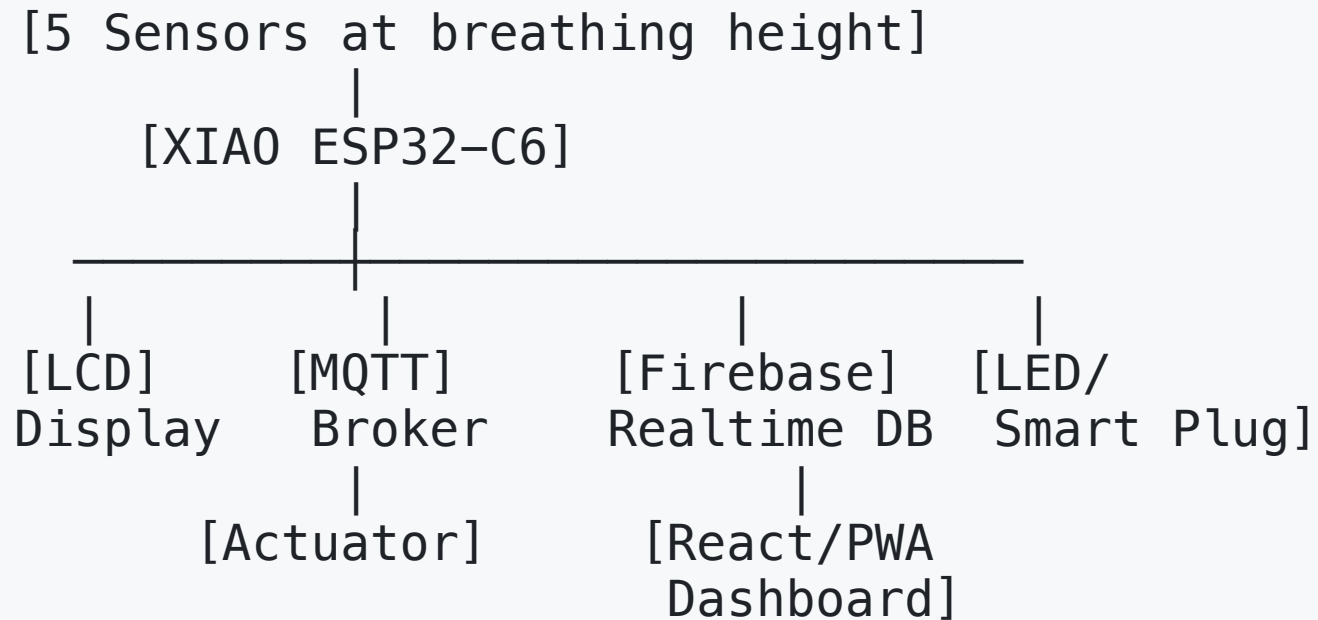
- CO2 above 1000 ppm causes measurable cognitive decline
- High PM2.5/NO2 increases sleep issues by **50–60%**
- People spend **90%** of their time indoors

Why This Hasn't Been Solved

Issue	Commercial Purifiers	This Project
Sensor location	Inside device (cleaned air)	Breathing zone (3–6 ft)
Parameters monitored	PM2.5 only	CO2, PM2.5, VOC, Temp, Humidity
Works with any purifier	No — brand locked	Yes — via smart plug
Data export	Closed system	Open — Firebase + GitHub
Cost	\$200–400 per room	\$88 total

My Solution

A 5-sensor IoT monitor placed where you breathe



Hardware — What I Built

Component	Model	Protocol	Status
Microcontroller	XIAO ESP32-C6	—	✓ Deployed
CO2 Sensor	MH-Z19C (NDIR)	UART	✓ Working
PM2.5 Sensor	DC01	UART	⚠ Intermittent
VOC Sensor	MS1100	Analog	✓ Working
Temp/Humidity	AHT10	I2C	✓ Working
Display	LCD1602	I2C	✓ Working
Actuator	LED (smart plug stand-in)	GPIO	✓ Working

Total cost: \$88

Sprint 1 — Foundation (Weeks 1–6)

-  GitHub repositories created and documented
-  Canvas project + progress pages live throughout semester
-  PPP presentation delivered (Week 4)
-  Arduino IDE + XIAO ESP32-C6 configured
-  Key discovery: XIAO uses **GPIO22/23** for I2C — not standard GPIO21/22
-  AHT10, LCD1602, MH-Z19C, MS1100 individually tested + verified
-  All sensors running **simultaneously** on a single board

LCD Screen 1 (3 sec):

Temp: 24.1 C

Hum: 34.5%

LCD Screen 2 (3 sec):

CO2: 1600 ppm

VOC: 1874

Sprint 2 — Intelligence (Weeks 7–14)

Weeks 7–8: DC01 PM2.5 connected · MQTT broker live · LED automation triggered by thresholds · Firebase Realtime Database logging every 10 seconds

Weeks 9–10: React/PWA dashboard deployed · Real-time charts via Recharts · Firebase `onValue` for instant updates · PWA installed on phone via manifest

Weeks 11–12: Sleep quality tracking interface built and deployed · 4+ weeks continuous data needs to be collected · Sensor health auto-detection added

Weeks 13–14: Moving average calibration applied · Checksum validation for PM2.5 sensor · Documentation complete · Final presentation + demo video

Automation Logic

The system automatically triggers the actuator based on sensor thresholds:

Condition	Action
CO2 > 1000 ppm	LED ON (smart plug would activate purifier)
PM2.5 > 35 $\mu\text{g}/\text{m}^3$	LED ON
VOC > 2500 (raw ADC)	LED ON
All readings normal	LED OFF

Protocol: MQTT over Wi-Fi — latency < 2 seconds

Demo: LED used as actuator stand-in. In real deployment, a Tasmota-flashed smart plug replaces the LED — same MQTT message, same automation logic, any purifier brand.

Cloud Dashboard — React + PWA

- Real-time sensor readings — Firebase `onValue` pushes every update instantly
- Historical charts for CO2, PM2.5, VOC, Temp, Humidity (1h / 24h / 7d / All Data)
- Sensor health detection — auto-flags stuck or disconnected sensors
- Threshold alert badges + alert history log
- LED control step chart — shows exact ON/OFF timing
- Sleep quality rating interface — logs nightly scores (1–10) to Firebase
- **PWA**: add to phone home screen from Safari or Chrome — no App Store needed

Demonstration

Demo Video

What the video shows:

1. Sensors reading live — CO2 rising in a closed room
2. CO2 threshold exceeded → LED turns on automatically
3. Firebase dashboard updates in real time
4. Historical chart and time range selection
5. PWA installed on phone — launching from home screen
6. Sleep quality log entry

Demo video: https://drive.google.com/drive/folders/1dxNxLwAFDM-qzTNDedeU8n3i4qP3a_3x?usp=sharing

Technology Stack

Hardware / Firmware

- XIAO ESP32-C6 · Arduino IDE · C++
- Libraries: Adafruit AHTX0, LiquidCrystal_I2C, MHZ19, PubSubClient
- Protocols: I2C, UART, Analog, MQTT

Backend / Cloud

- Firebase Realtime Database — real-time time-series data storage
- MQTT (HiveMQ public broker) — actuator control, < 2s latency

Frontend / Mobile

- React + Vite + PWA — single codebase for desktop and mobile
- Recharts — real-time sensor data visualization

Learning with AI

Topic 1: Multi-Sensor Integration & Calibration

Used AI to understand XIAO-specific pin mapping, debug I2C bus sharing, implement UART protocol for MH-Z19C, and apply moving average calibration

Topic 2: Mobile App Development with React/PWA

Used AI to scaffold React dashboard components, implement Firebase Realtime Database listeners, structure Recharts time-series visualization, and build PWA manifest

Primary AI tool: Claude (Anthropic) — used for concept explanations, code debugging, protocol understanding, and documentation throughout both sprints

What I Would Do Differently

1. **Start data collection in Week 4** — more weeks of data enables future ML sleep quality correlation
2. **Swap DC01 for PMS5003** — PMS5003 uses laser scattering for PM2.5; more accurate and more reliable UART communication
3. **Real Tasmota smart plug** — The LED demo proves automation logic; a smart plug adds real appliance control with zero code changes
4. **Multi-room support** — A second XIAO unit would allow comparing bedroom vs. living room air quality in real time

Results Summary

-  **Problem solved:** Sensors at breathing zone, not inside purifier
-  **5 sensors running 24/7** for 4+ weeks uninterrupted
-  **Automated actuator control** via CO2, PM2.5, and VOC thresholds over MQTT
-  **Public PWA dashboard** — real-time + historical data, installable on any phone
-  **Sleep quality logger** — nightly scores stored in Firebase alongside sensor data
-  **Total cost: \$88** vs. \$200–400 commercial alternatives
-  **Open source** — all code on GitHub, live at smart-home-health-monitor.web.app

Repository Links

Project Code & Documentation

<https://github.com/shresthamausam07/smart-home-health-monitor>

Learning with AI Documentation

<https://github.com/shresthamausam07/mausam-shrestha-learning-with-ai>

Canvas Project Page

<https://nku.instructure.com/courses/88152/pages/individual-project-mausam-shrestha>

Canvas Progress Page

<https://nku.instructure.com/courses/88152/pages/individual-progress-mausam-shrestha>

Live Dashboard

<https://smart-home-health-monitor.web.app>

Thank You

Any Questions?

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"The goal was to know what you're actually breathing — not what the purifier thinks you're breathing."